SUMMARY ALARM (FAIL)	EFI-1 SUPERVISORY RESET	EFI-2 SUPERVISORY RESET	EFI-1 240VAC/AC RESET L | N	EFI-2 240VAC/AC RESET L | N

INTERFACE TO DC CHARGER

*REFER TO 24VDC CHARGER DRAWING

-X4 / C04 C NO

INTERFACE TO EFI

REFER TO EMG EASI-R A1-1908/C3

11 12 11 12

REFER TO EMG EASI-R A1-1908/C7

13 14 13 14

L1 L2 R3 R4 R5 R6 H9 H10 H11 H12

-XC 1 2 -XC 3 4 -XC 5 6 -XH 7 8

L1 L2 R3 R5 R6 (-ve)

REFER TO CSU-SCA-RCB-010

*RCB

BY OTHERS INTERFACE TO RTU

-DI 37 26 -DO 21 -DO 23 40

-DI REFER TO DRAWING CSU-SCA-RTU-03-008-01
-DO REFER TO DRAWING CSU-SCA-RTU-03-009-01

NEW

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Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019

Title: COMPACT SUBSTATION UNIT SCADA READY 500kVA & 1000kVA

Description: REMOTE CONTROL BOX CONTROL AND INDICATION CIRCUIT COMMON SIGNAL

MATERIAL Type / Code

COATING Type / Code

GENERAL TOLERANCE: XX ±05, XX-XXXX ±10, UNLESS SPECIFIED

PROJECTION

INDKOM ENGINEERING SDN BHD

Approved by: EN JAMAL
Checked by: EN ALI, EN HAFIZ
Drawn by: SARAH

Client: TENAGA NASIONAL BERHAD FOR SGP APPLICATION

Dwg No: CSU-SCA/RD-RCB-008

Sheet 9 / 10

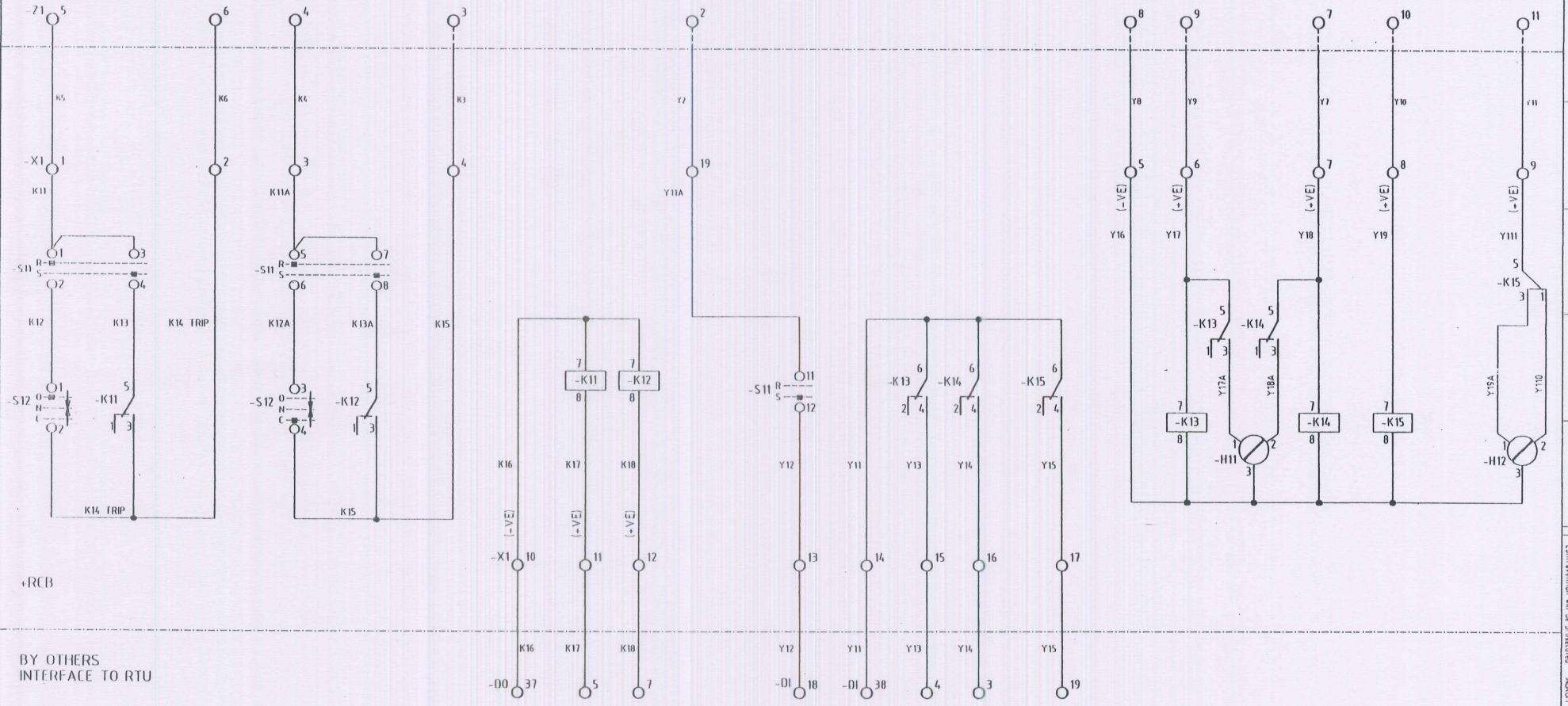
WM Code

Tender No

Dept	Year	Scale	Units
ENG	2024		mm

OPEN COMMON	SUPERVISORY OPEN	REMOTE OPEN	CLOSE COMMON	SUPERVISORY CLOSE	REMOTE CLOSE	COMMON -VE	SUPERVISORY OPEN | CLOSE	COMMON +VE	SUPERVISORY STATUS	COMMON +VE	LBS 1 STATUS CLOSE | OPEN	EARTH SWICTH STATUS CLOSE	COMMON -VE	LBS 1 STATUS CLOSE | OPEN	EARTH SWITCH STATUS CLOSE | OPEN

INTERFACE TO RMU REFER TO DRAWING CSU-SCA-RMU-001
LEGEND : 7H TO 10H



BY OTHERS
INTERFACE TO RTU

-DI REFER TO DRAWING CSU-SCA-RTU-03-008-01
-DO REFER TO DRAWING CSU-SCA-RTU-03-009-01

Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019

Title	COMPACT SUBSTATION UNIT SCADA READY 500kVA & 1000kVA
Description	REMOTE CONTROL BOX CONTROL AND INDICATION CIRCUIT INCOMER FEEDER -LBS 1

INDKOM ENGINEERING SDN BHD

Approved by	EN JAMAL
Checked by	EN ALI EN HAFIZ
Drawn by	SARAH

Client: TENAGA NASIONAL BERHAD FOR SGP APPLICATION

Dwg No	CSU-SCA/RD-RCB-009
Sheet	10 / 10
Dept	ENG
Year	2024
Scale	NTS
Units	mm
IM Code	-
Tender No	

	Type		Type	GENERAL TOLERANCE	PROJECTION
MATERIAL	Code	COATING	Code	XX ±0.5 XX-XXXX ±0.10 UNLESS SPECIFIED	

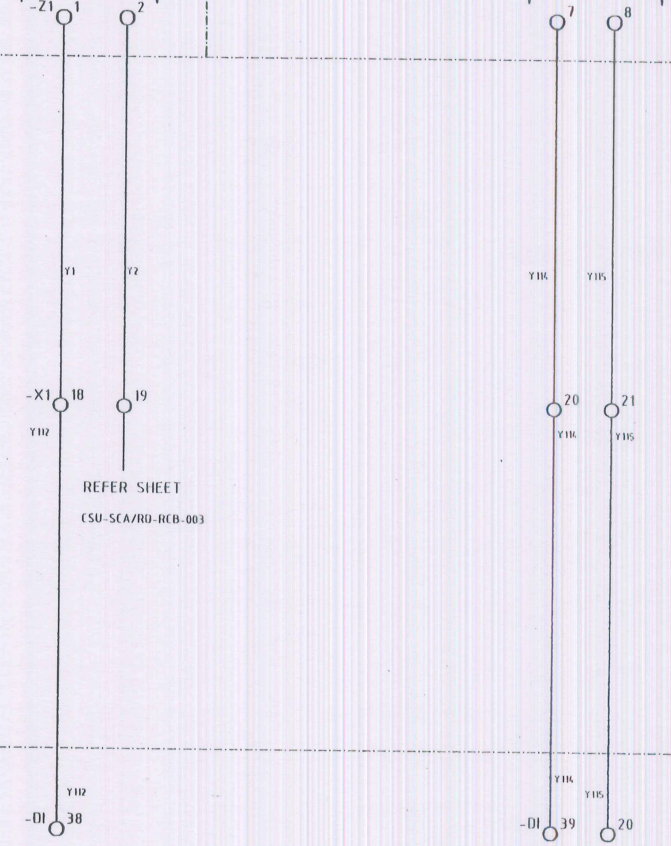
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FORM NO. QA-F-01-121

RMU REMOTE STATUS LBS 1	EFI 1 STATUS
INTERFACE TO RMU	INTERFACE TO EFI

REFER TO DRAWING
CSU-SCA-RMU-001
LEGEND : 10H

REFER TO
EMG EASI-R A1-1908/B3



-DI REFER TO DRAWING CSU-SCA-RTU-03-008-01

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Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019

Title	COMPACT SUBSTATION UNIT SCADA READY 500kVA & 1000kVA
Description	REMOTE CONTROL BOX CONTROL AND INDICATION CIRCUIT INCOMER FEEDER -LBS 1

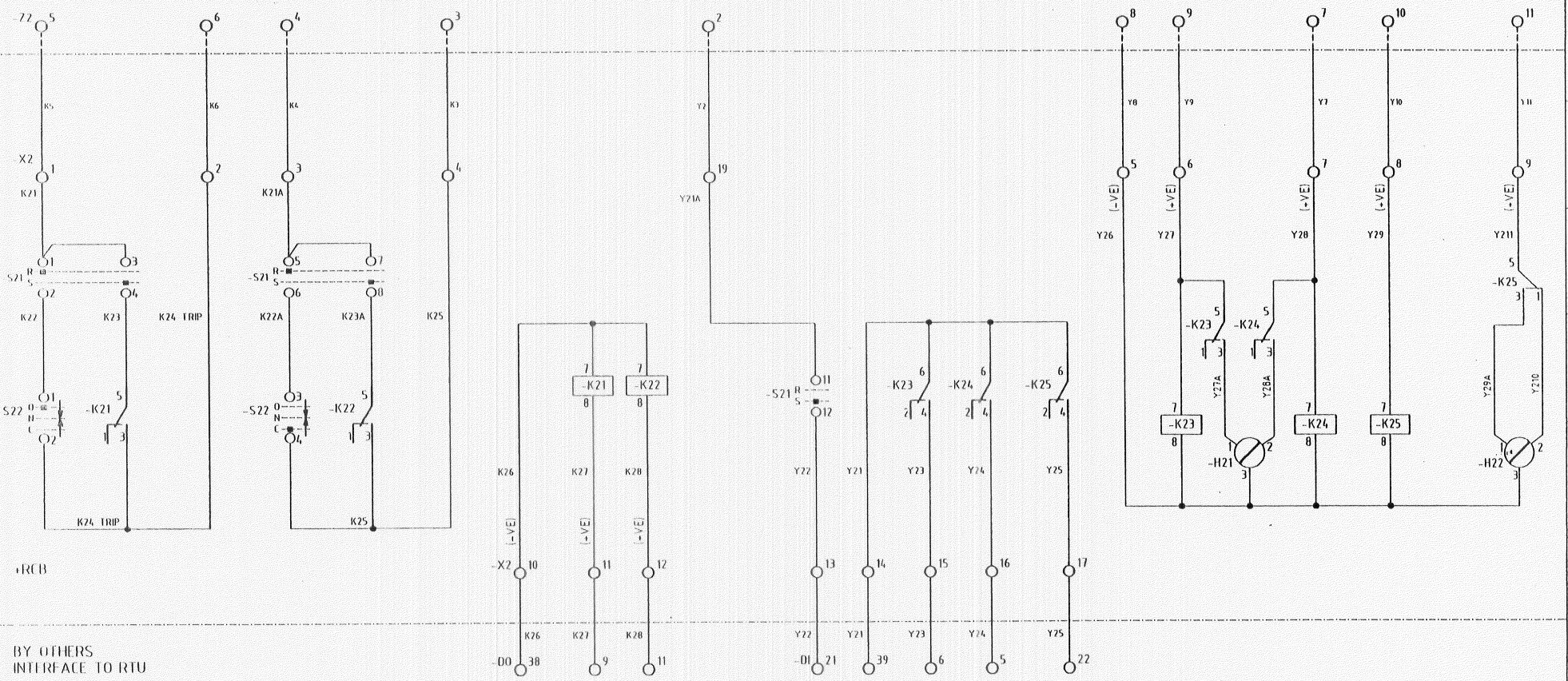
INDKOM ENGINEERING SDN. BHD.

MATERIAL	Type / Code
COATING	Type / Code
GENERAL TOLERANCE	XX ±05 XX-XXXX ±10 UNLESS SPECIFIED
PROJECTION	
Approved by	EN JAMAL
Checked by	EN ALI EN HAFIZ
Drawn by	SARAH
Client	TENAGA NASIONAL BERHAD FOR SGP APPLICATION
Dwg No	CSU-SCA/RD-RCB-010
Sheet	11 / 18
IM Code	
Contract No	
Dept	ENG
Year	2024
Scale	NTS
Units	mm

FORM NO QA-F-01 (2)

OPEN COMMON	SUPERVISORY OPEN	REMOTE OPEN	CLOSE COMMON	SUPERVISORY CLOSE	REMOTE CLOSE	COMMON -VE	SUPERVISORY OPEN | CLOSE	COMMON +VE	SUPERVISORY STATUS	COMMON +VE	LBS 2 STATUS CLOSE | OPEN	EARTH SWICTH STATUS CLOSE	COMMON -VE	LBS 2 STATUS CLOSE | OPEN	EARTH SWITCH STATUS CLOSE | OPEN

INTERFACE TO RMU REFER TO DRAWING CSU-SCA-RMU-001
LEGEND : 5H TO 7H



BY OTHERS
INTERFACE TO RTU

-DI REFER TO DRAWING CSU-SCA-RTU-03-008-01
-DO REFER TO DRAWING CSU-SCA-RTU-03-009-01

Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 25 OCT 2019			30	OCT	2019

Title: COMPACT SUBSTATION UNIT
SCADA READY 500kVA & 1000kVA

Description: REMOTE CONTROL BOX CONTROL AND INDICATION
CIRCUIT INCOMER FEEDER -LBS 2

INDKOM ENGINEERING SDN BHD

Approved by :	EN JAMAL
Checked by :	EN ALI EN HAFIZ
Drawn by :	SARAH

MATERIAL: Type / Code
COATING: Type / Code
GENERAL TOLERANCE: X X ±0.5, XX-XXXX ±0.10, UNLESS SPECIFIED
PROJECTION

Client: TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

IM Code :
Tender No :

Dwg No : CSU-SCA/RD-RCB-011
Sheet 12 of 18
Dept: ENG | Year: 2024 | Scale: Draft

FORM NO QA-F-01-12C

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RMU
REMOTE STATUS
LBS 2

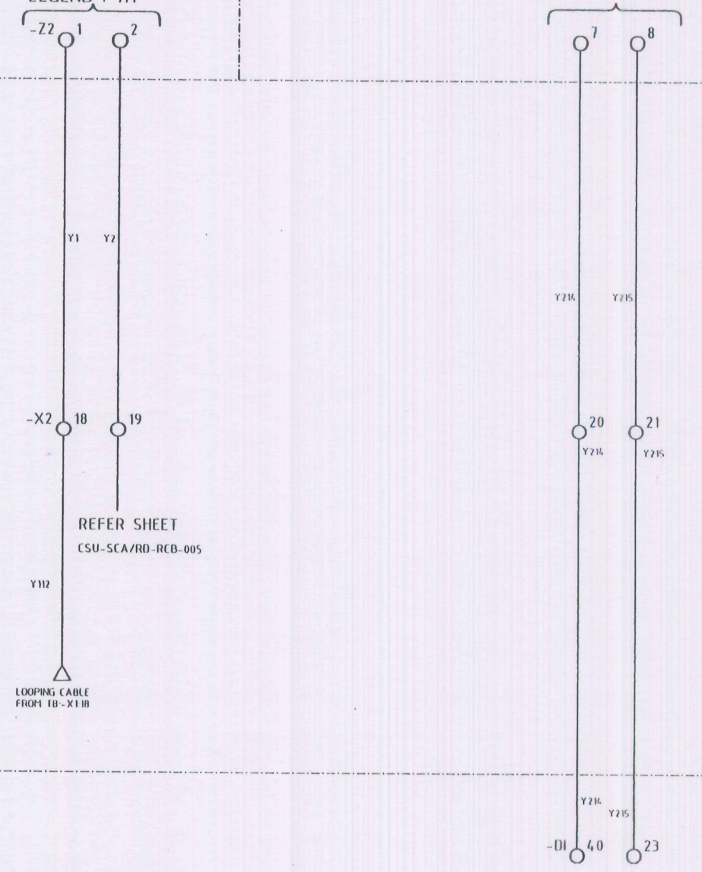
EFI 2
STATUS

INTERFACE TO RMU

REFER TO DRAWING
CSU-SCA-RMU-001
LEGEND : 7H

INTERFACE TO EFI

REFER TO
EMG EASI-R A1-1908/B3



BY OTHERS
INTERFACE TO RTU

-DI REFER TO DRAWING CSU-SCA-RTU-03-008-01

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Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019

Title : COMPACT SUBSTATION UNIT
SCADA READY 500kVA & 1000kVA

Description : REMOTE CONTROL BOX CONTROL AND INDICATION
CIRCUIT INCOMER FEEDER -LBS 2

INDKOM ENGINEERING SDN BHD

MATERIAL Type : Code :

COATING Type : Code :

GENERAL TOLERANCE
XX ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

Approved by : EN JAMAL
Checked by : EN ALI
EN HAFIZ
Drawn by : SARAH

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

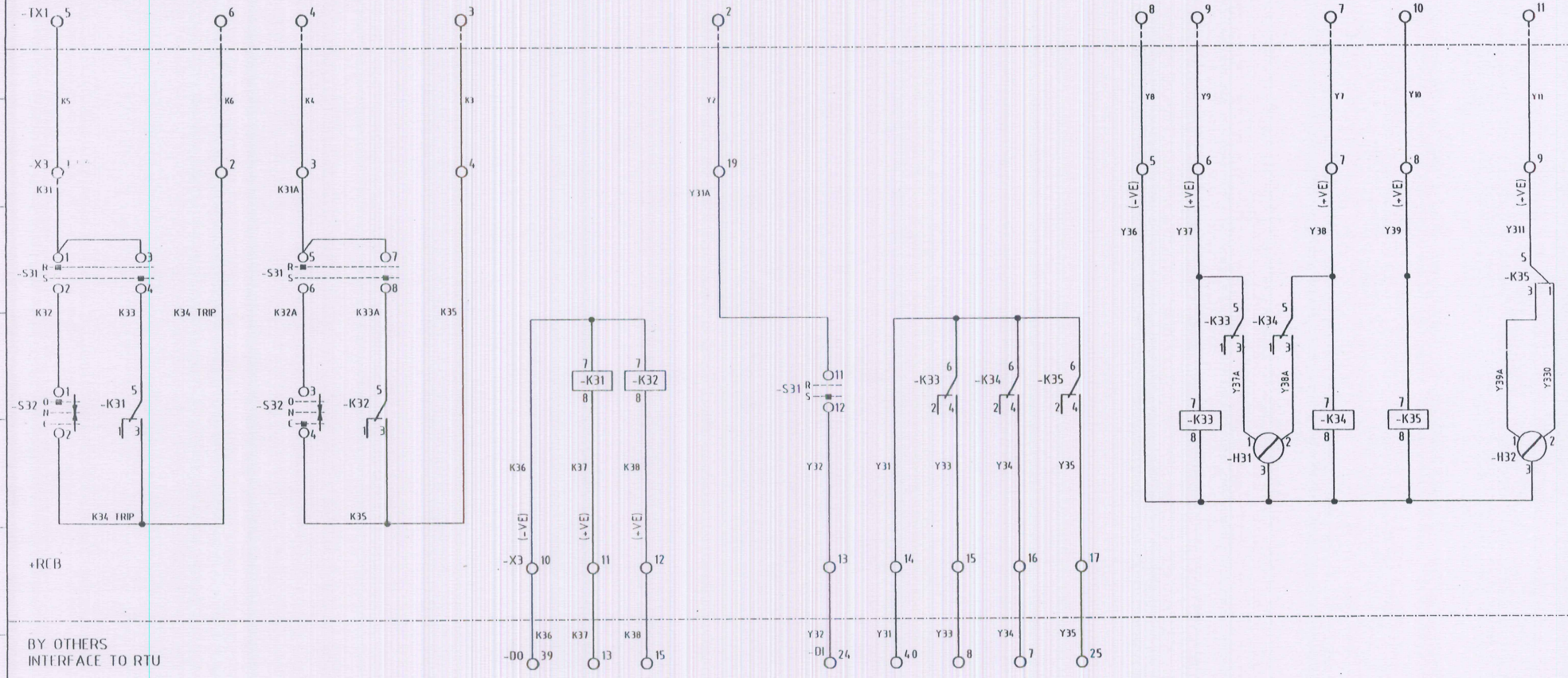
Dwg No : CSU-SCA/RD-RCB-012
Sheet 13 / 18

HH Code :
Contract No :
Dept. ENG
Year 2024
Scale -
Units mm

FORM NO. QA-F-01-121

OPEN COMMON	SUPERVISORY OPEN	REMOTE OPEN	CLOSE COMMON	SUPERVISORY CLOSE	REMOTE CLOSE	COMMON -VE	SUPERVISORY OPEN | CLOSE	COMMON +VE	SUPERVISORY STATUS	COMMON +VE	TS STATUS CLOSE | OPEN	EARTH SWICTH STATUS CLOSE	COMMON -VE	TS STATUS CLOSE | OPEN	EARTH SWITCH STATUS CLOSE | OPEN

INTERFACE TO RMU REFER TO DRAWING CSU-SCA-RMU-001
LEGEND : 1H TO 4H



BY OTHERS
INTERFACE TO RTU

-DI REFER TO DRAWING CSU-SCA-RTU-03-008-01
-DO REFER TO DRAWING CSU-SCA-RTU-03-009-01

Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019

Title	COMPACT SUBSTATION UNIT SCADA READY 500kVA & 1000kVA
Description	REMOTE CONTROL BOX CONTROL AND INDICATION CIRCUIT INCOMER FEEDER -TS

INDKOM ENGINEERING SDN BHD

Approved by	EN JAMAL
Checked by	EN ALI EN HAFIZ
Drawn by	SARAH

MATERIAL	Type / Code
COATING	Type / Code :
GENERAL TOLERANCE	XX ±05 XX-XXXX ±10 UNLESS SPECIFIED
PROJECTION	
Client	TENAGA NASIONAL BERHAD FOR SGP APPLICATION
Dwg No	CSU-SCA/RD-RCB-013
Sheet	14 / 18
H1 Code	
Tender No	
Dept	ENG
Year	2024
Scale	thats mm

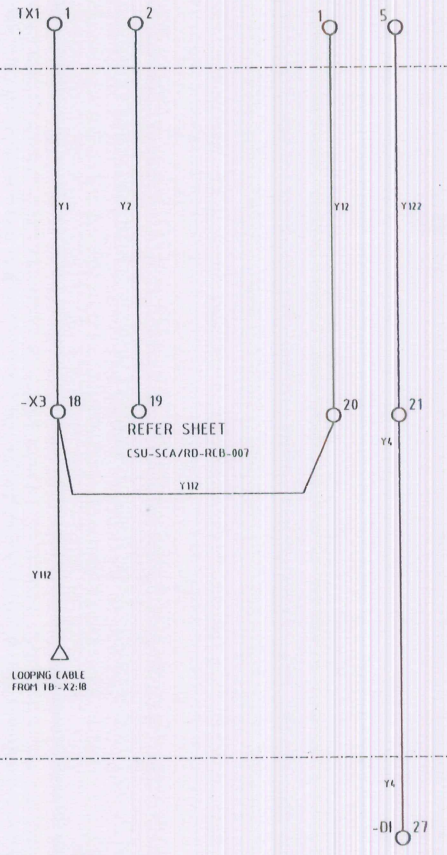
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FORM NO: QA-F-01-171

RMU LOCAL/REMOTE TS	RMU MCB TRIP/OFF

INTERFACE TO RMU REFER TO DRAWING CSU-SCA-RMU-001
LEGEND : 4H

N/C CONTACT RELAY FOR MCB TRIP



+RCB

BY OTHERS
INTERFACE TO RTU

-D1 REFER TO DRAWING CSU-SCA-RTU-03-008-01

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Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019

Title	COMPACT SUBSTATION UNIT SCADA READY 500kVA & 1000kVA
Description	REMOTE CONTROL BOX CONTROL AND INDICATION CIRCUIT INCOMER FEEDER -TS

INDKOM ENGINEERING SDN BHD

Approved by	EN JAMAL
Checked by	EN ALI EN HAFIZ
Drawn by	SARAH

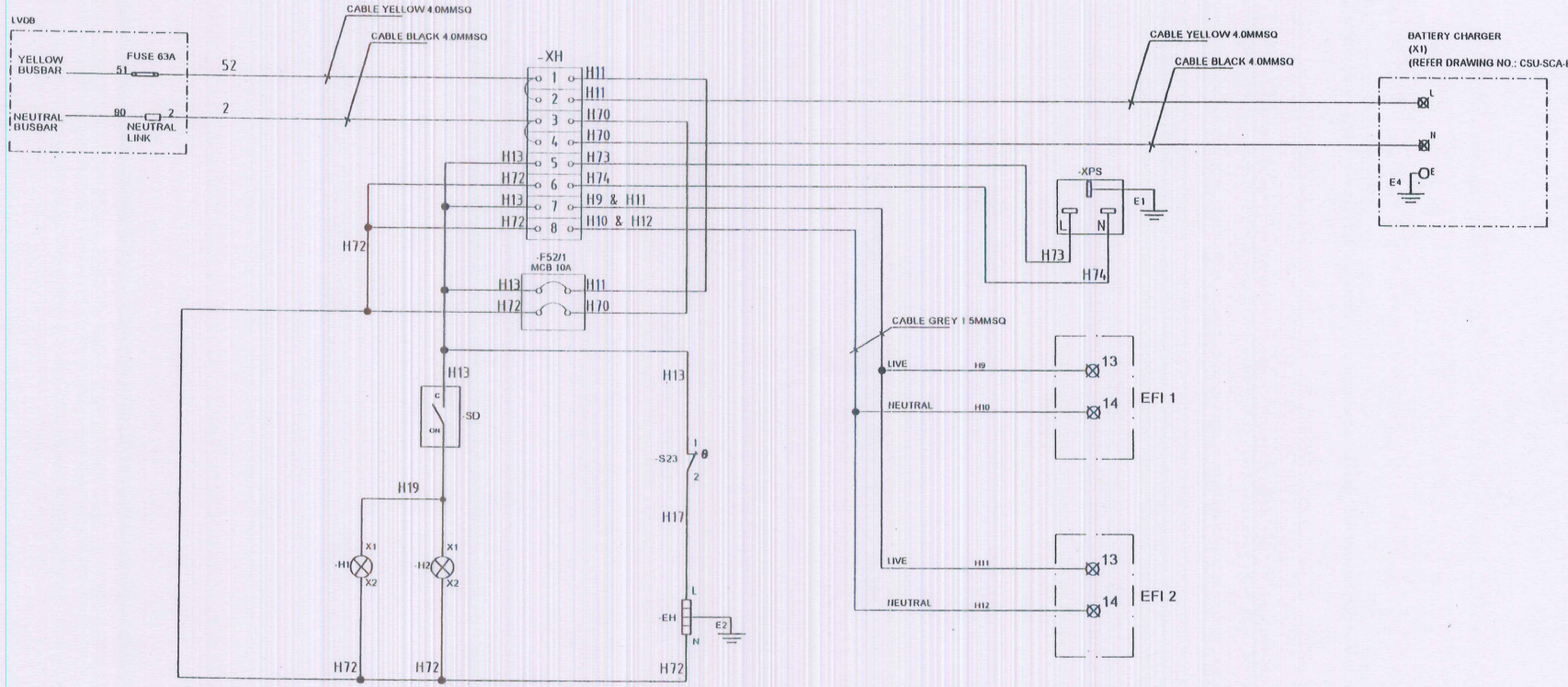
MATERIAL	Type / Code
COATING	Type / Code
GENERAL TOLERANCE	XX ±05 XX.XXX ±10 UNLESS SPECIFIED
PROJECTION	

Client: TENAGA NASIONAL BERHAD FOR SGP APPLICATION

Dwg No: CSU-SCA/RD-RCB-014 | Sheet 15 / 18

IM Code	Contract No	Dept	Year	Scale	Units
		ENG	2024	-	mm

FORM NO: QA-F-01.121



Rev	Reason	X	Y	Day	Month	Year
01	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 28 AUG 2024			29	AUG	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 08 OCT 2020			23	NOV	2020
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019

Title: COMPACT SUBSTATION UNIT
SCADA READY 500kVA & 1000kVA

Description: AC CIRCUIT

INDKOM

Approved by	EN JAMAL
Checked by	EN ALI EN HAFIZ
Drawn by	SARAH

Client: TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No: CSU-SCA/RD-RCB-015

Sheet 16 of 16

Dept	Year	Scale	Units
ENG	2024	-	mm

MATERIAL	Type / Code	COATING	Type / Code	GENERAL TOLERANCE	PROJECTION
				X.X ±0.5 XX-XXXX ±1.0 UNLESS SPECIFIED	

HT Code

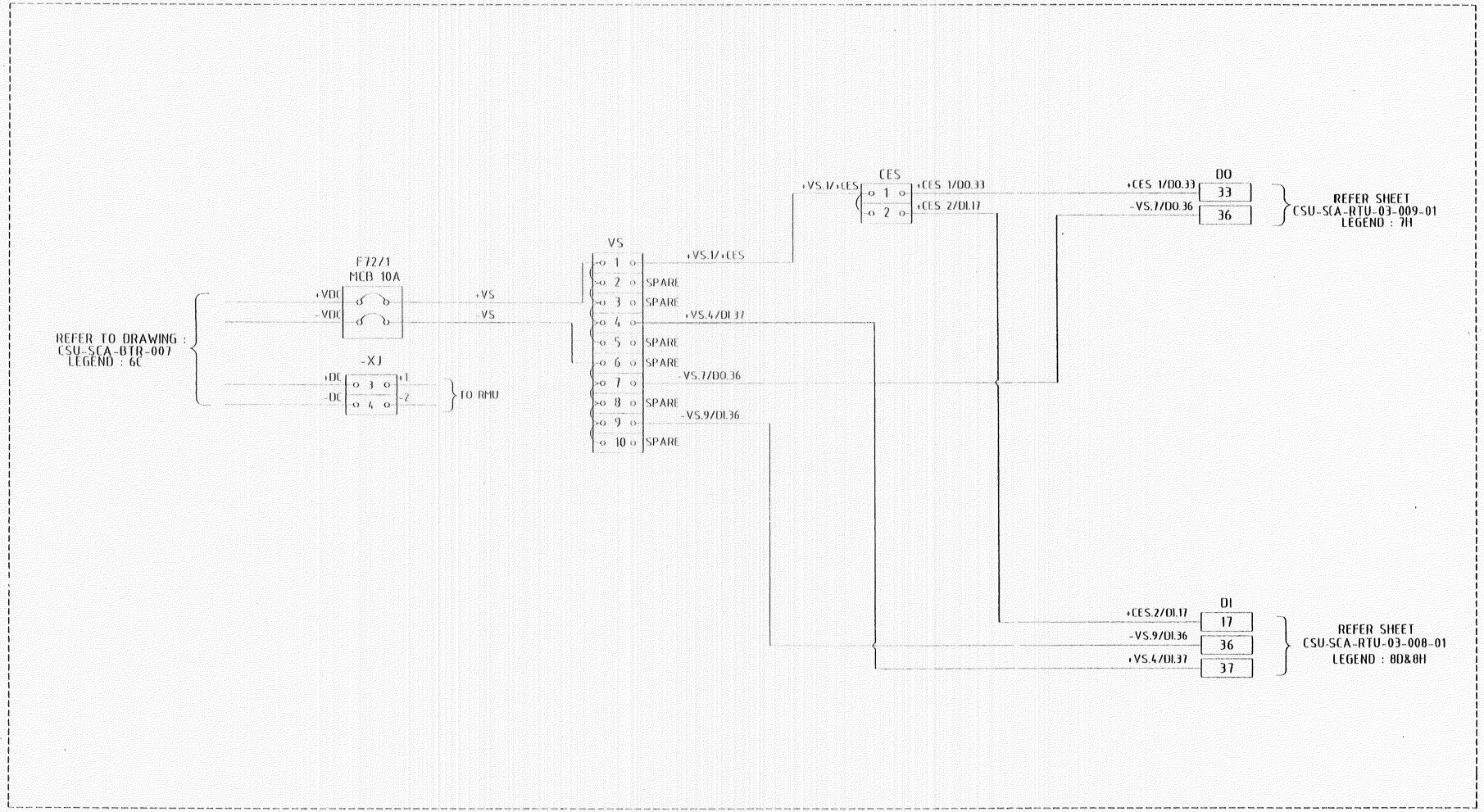
Contract No

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FORM NO. QA-F-01-121

SCADA READY

RTU COMPARTMENT



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Rev	Reason	X	Y	Day	Month	Year
01	UPDATE SGP			29	AUG	2024
00	INITIAL RELEASE			02	DEC	2019

Title: COMPACT SUBSTATION UNIT SCADA READY 500kVA & 1000kVA

Description: DC DISTRIBUTION WIRING DIAGRAM -2

INDKOM ENGINEERING SDN BHD

Approved by	EN. JAMAL
Checked by	EN. ALI / EN. HAFIZ
Drawn by	SARAH

MATERIAL: Type / Code

COATING: Type / Code

GENERAL TOLERANCE: XX ±0.5; XX-XXXX ±1.0; UNLESS SPECIFIED

PROJECTION

Client: TENAGA NASIONAL BERHAD FOR SGP APPLICATION

IPI Code

Tender No

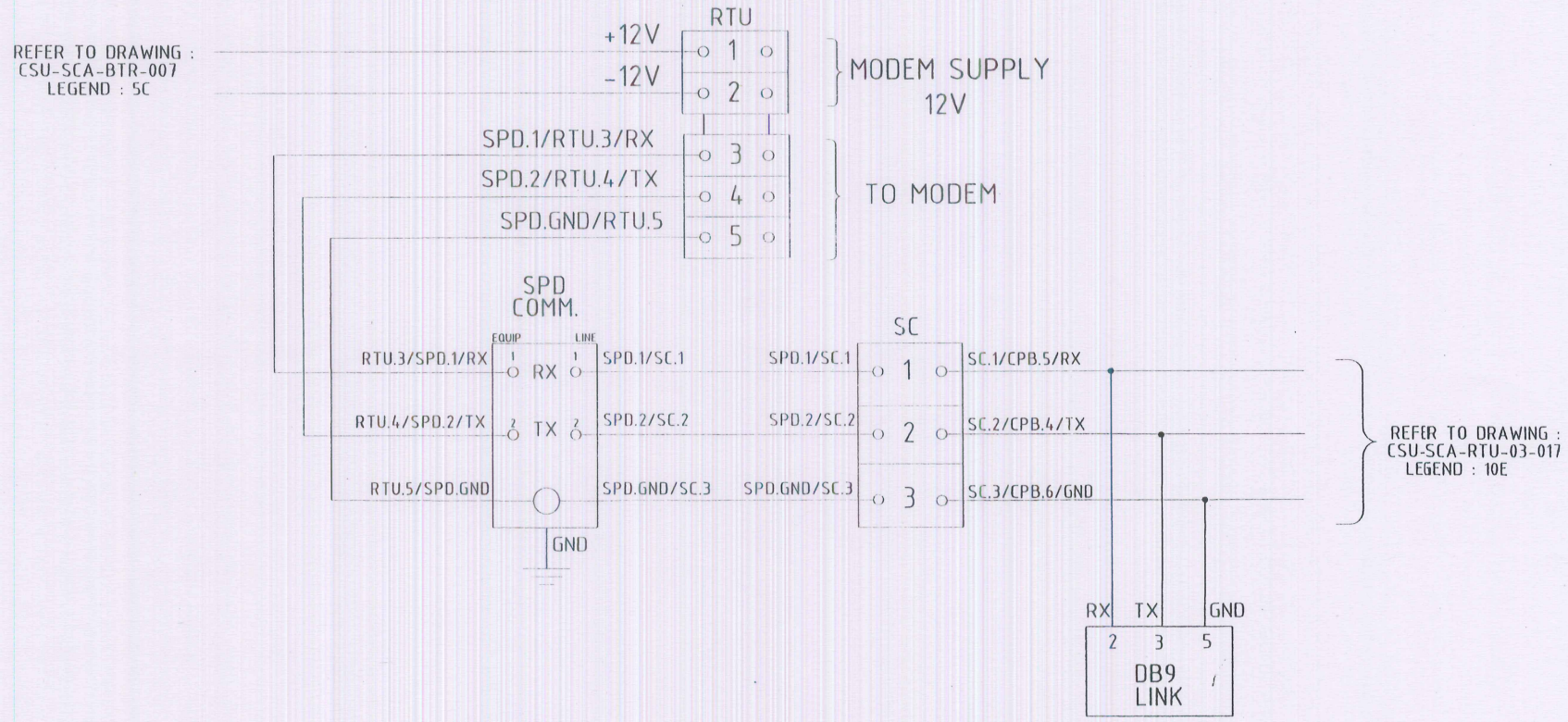
Dwg No: CSU-SCA/RD-RCB-016-02

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Dept	Year	Scale	Units
ENG	2024		mm

FORM NO: QA-F-01-121

EQUIPPED



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Rev	Reason	X	Y	Day	Month	Year
01	UPDATE SGP			29	AUG	2024
00	INITIAL RELEASE			02	DEC	2019

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : DC DISTRIBUTION WIRING DIAGRAM -1

MATERIAL Type / Code

COATING Type / Code

GENERAL TOLERANCE
XX ±0.5
XX.XXXX ±0.10
UNLESS SPECIFIED

PROJECTION

INDKOM
TEKNOLOGI SDN BHD

Approved by : EN. JAMAL
Checked by : EN. ALI / EN. HAFIZ
Drawn by : SARAH

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Org No : CSU-SCA-RCB-016-01

Sheet 11 of 18

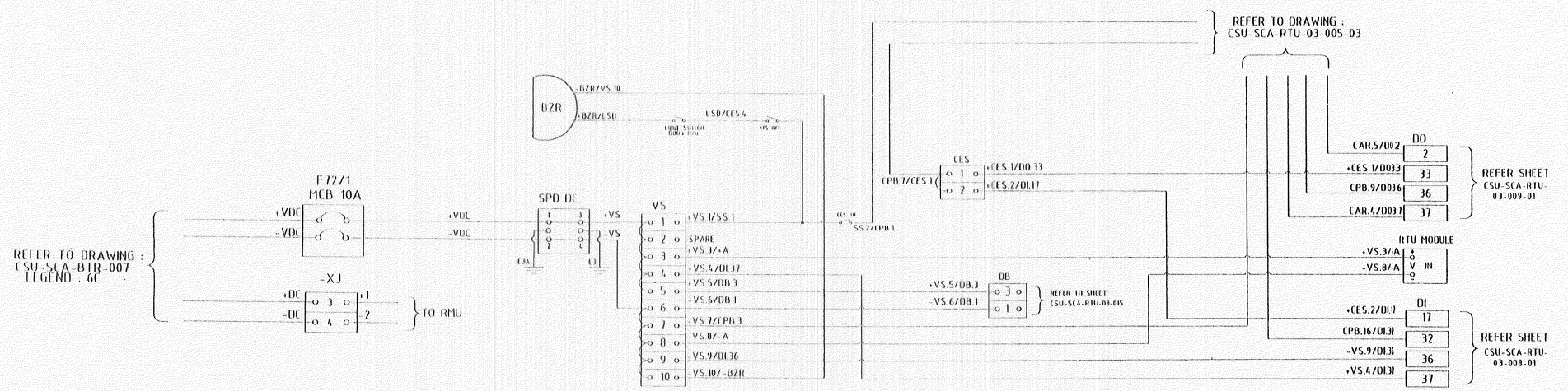
IH Code

Tender No

Dept	Year	Scale	Units
ENG	2024	.	mm

FORM NO. DA-F-01-12

EQUIPPED



Rev	Reason	X	Y	Day	Month	Year
01	UPDATE SLD			29	AUG	2024
00	INITIAL RELEASE			02	DEC	2019

Title: COMPACT SUBSTATION UNIT SCADA EQUIPPED 500kVA & 1000kVA

Description: DC DISTRIBUTION WIRING DIAGRAM -2

INDKOM ENGINEERING SDN. BHD.

Approved by	EN. JAMAL
Checked by	EN. ALI EN. HAFIZ
Drawn by	SARAH

Client: TENAGA NASIONAL BERHAD FOR SGP APPLICATION

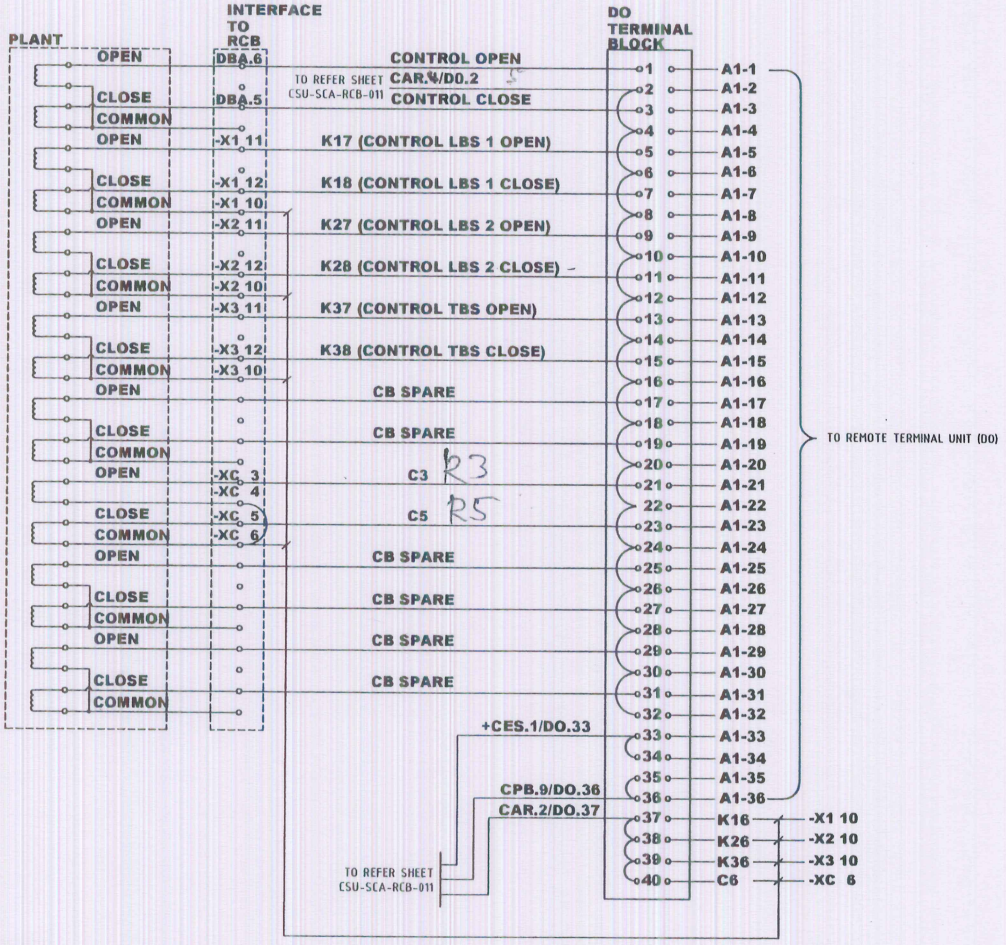
Dwg No: CSU-SCA-RCB-016-02

Sheet 10 of 16

MATERIAL	Type / Code	COATING	Type / Code
GENERAL TOLERANCE	XX ±0.5 XX.XXXX ±0.10 UNLESS SPECIFIED	PROJECTION	
IM Code		Tender No	
Dept	ENG	Year	2024
Scale	-	Units	mm

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Rev.	Reason	X	Y	Day	Month	Year
03	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 25 OCT 2023			20	MAY	2024
02	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019
01	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 03 JUL 2018			11	JUL	2018

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : PROVISION TO RTU WIRING
DIGITAL OUTPUT WIRING DIAGRAM

INDKOM ENGINEERING SDN BHD

Approved by :	EN. JAMAL
Checked by :	EN. ALI EN. HAFIZ
Drawn by :	SHAHARIN

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No : CSU-SCA-RCB-012

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MATERIAL	Type : -	Code : -
COATING	Type : -	Code : -

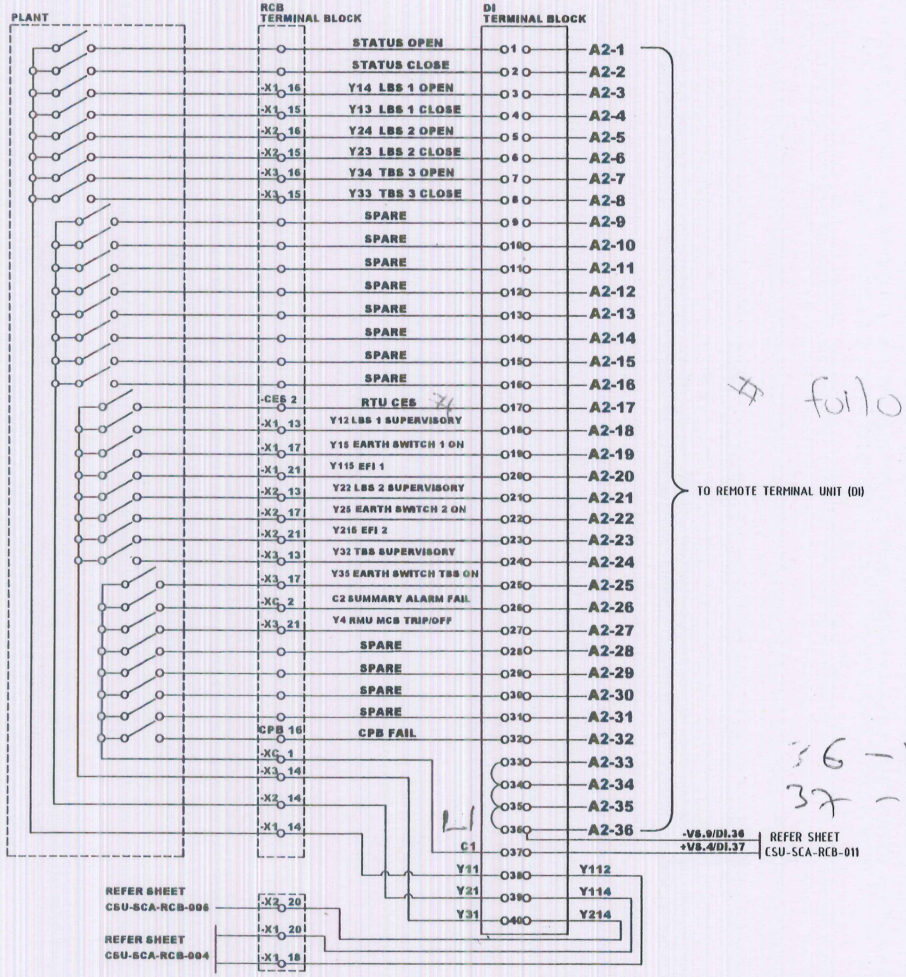
GENERAL TOLERANCE
XX ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

IM Code : -
Contract No : -

Dept.	Year	Scale	Units
ENG	2024	-	mm

FORM NO: QA-F-01.12F



Rev	Reason	X	Y	Day	Month	Year
04	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 25 OCT 2023			20	MAY	2024
03	DRAWING UPDATE AS PER TNB COMMENT ON SGP APPLICATION 24 OCT 2019			30	OCT	2019
02	DRAWING UPDATE AS PER TNB COMMENT ON FUNCTIONAL TEST 03 JUL 2018			11	JUL	2018

Title : COMPACT SUBSTATION UNIT
SCADA EQUIPPED 500kVA & 1000kVA

Description : PROVISION TO RTU WIRING
DIGITAL INPUT WIRING DIAGRAM

INDKOM ENGINEERING SDN. BHD.

Approved by :	EN. JAMAL
Checked by :	EN. ALI EN. HAFIZ
Drawn by :	SHAHARIN

Client : TENAGA NASIONAL BERHAD
FOR SGP APPLICATION

Dwg No : CSU-SCA-RCB-013

Sheet 14 of 41

Dept	Year	Scale	Units
ENG	2024	-	mm

IM Code :

Contract No. :

MATERIAL Type : - Code : -

COATING Type : - Code : -

GENERAL TOLERANCE
XX ±0.5
XX-XXXX ±1.0
UNLESS SPECIFIED

PROJECTION

FORM NO: QA-F-01-12F

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